
Problem 8.25. An undirected graph is ***bipartite*** if its nodes may be divided into two sets so that all edges go from a node in one set to a node in the other set. Show that a graph is bipartite if and only if it doesn't contain a cycle that has an odd number of nodes. Let $BIPARTITE = \{\langle G \rangle \mid G \text{ is a bipartite graph}\}$. Show that $BIPARTITE \in \mathrm{NL}$.

Part a. Show that a graph is bipartite if and only if it doesn't contain a cycle that has an odd number of nodes.

Proof. For the forward direction, let G be a bipartite graph and show that G doesn't contain a cycle that has an odd number of nodes. The proof is by contradiction. Assume that G contains a a cycle $n_1, n_2, \ldots, n_{k-1}, n_k$ that has an odd number of nodes. Then, it must be the case that nodes n_1 and n_{k-1} are assigned to different sets because every two consecutive nodes in the sequence $n_1, n_2, \ldots, n_{k-1}, n_k$ are adjacent to each other. But this implies that node n_k cannot be assigned to either first or second set as it is adjacent to both nodes n_1 and n_{k-1}, which is a contradiction.

To prove the other direction of the theorem, □

Part b. Show that $BIPARTITE \in \mathrm{NL}$.

Proof. □